

# Xiaorui Gu

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## Education

### Northwestern University

M.S. in Electrical Engineering

Advisors: Prof. Ed Colgate & Prof. Randy Freeman

Evanston, IL  
Sep 2025 – Jun 2027 (Expected)

### University of Illinois Urbana-Champaign

B.S. in Mechanical Engineering, Honors

Urbana, IL  
Aug 2021 – May 2025

## Research Experience

### HAND ERC (Human Augmentation via Dexterity Engineering Research Center)

Aug 2025 – Present

Northwestern University

- Developing a multimodal tactile perception system for a high-precision contact-rich task.
- Prototyped a contact-based tactile sensing e-skin with high durability.

### Mobility and Fall Prevention Research Laboratory

Jun 2024 – Mar 2025

Advisor: Prof. Manuel E. Hernandez, University of Illinois

- Developed a piezoresistive pressure sensor array, achieving high sensitivity and portability for human walking gait analysis and rehabilitation.

### Bahl Research Group

Aug 2023 – Dec 2024

Advisor: Prof. Gaurav Bahl, University of Illinois

- Designed and constructed experimental setups for a fully levitated 6-DOF system for low-frequency signal generation and inertial sensing experiments.
- Designed and fabricated custom mechanical and electronic hardware using CAD, CNC machining, laser cutting, waterjet cutting, 3D printing, and PCB assembly.

### Garg Group

May 2023 – Dec 2024

Advisor: Prof. Nishant Garg, University of Illinois

- Prototyped a temperature-humidity monitoring sensor array with Arduino and ESP8266, integrating wireless data collection with Firebase.
- Investigated monocular vision pipelines using OpenCV and Open3D for 3D reconstruction and concrete flow measurement.

## Publication

Gu, X., Gupta, P., Liu, J., Zhou, H., Cisto, B., Khan, M. A., Mason, S., Motl, R., Sebastiao, E., & Hernandez, M. E. *Intelligent Square Stepping Exercise System for Cognitive-Motor Rehabilitation in Older Adults with Multiple Sclerosis. Proceedings of the 2025 Design of Medical Devices Conference*, Minneapolis, MN.

## Selected Projects

### Autonomous Driving with Static & Dynamic Obstacle Avoidance

Jan 2025 – May 2025

Team: SafeTaxi, ECE484: Principles of Safe Autonomy, UIUC

- Developed lane following and obstacle avoidance algorithms using LiDAR and camera perception on an NVIDIA Jetson.

### Object Recognition on Mobile DSP

Aug 2024 – Dec 2024

Team: UIUC Campus Tour Application, ECE420: Embedded DSP Systems, UIUC

- Built an Android mobile app for landmark classification using SIFT feature extraction, K-means clustering, and SVM classification.

## Skills

**Programming** - Python, C++, Racket, MATLAB, Linux, Git

**CAD** - KiCAD, Autodesk Fusion 360, Onshape

**Libraries & Frameworks** - OpenCV, PyTorch, ROS2, Gazebo

**Embedded** - Teensy, Arduino, ESP32, Raspberry Pi, NVIDIA Jetson Nano